

1. INTRODUCTION - PROJECT PLANNING:

PocketSmart AI project-a success-a mudikka planning romba mukkiyam. Planning illama code ezhuthina confusion varum. Athunala namma Agile methodology follow pannom.

A) Project Timeline (4 Weeks Sprint): Week 1 - Project Setup & Authentication Module: VSCode-la folder create, Flask setup, requirements.txt login/register with JWT. Video-la neenga first 2 mins itha than pannuneenga.

Week 2 - Core Planner Logic: Home Planner, Party Planner, Jewelry Planner-ku thani thani Python function. Budget split logic. Prompt engineering start.

Week 3 - Gemini AI Integration & Frontend: Gemini 1.5 Flash API key vangarthu, app.py la integrate panrathu, HTML/CSS/JS dashboard create panrathu. Image upload feature for jewelry.

Week 4 - Testing, UI Polish, Documentation & Demo: Manual testing - pocketsmartai@gmail.com login test, 10k budget test, 40 guests test, saree image upload test. Bug fix: 10k ku 33k kaamichathu. README & 8 phases PDF ready.

B) Task Breakdown: T1: Environment Setup - Python, Flask, VSCode. T2: Auth Module - Login, Register, JWT, Security. T3: Planner Module - Home, Party, Jewelry prompt. T4: AI Integration - Gemini API, Vision API. T5: Frontend Integration - Dashboard, Cards, Fetch API. T6: Testing & Deployment - Unit test, Live demo.

C) Resource Planning: Single Developer, Laptop with VSCode, Gemini API Free Tier, Flask free, No budget needed.

D) Risk Planning: Risk 1: Gemini API limit - Solution: Error handling add pannom. Risk 2: Budget calculation thappu - Solution: Prompt-la 'Strictly within budget' nu mention. Risk 3: Frontend-backend connection fail - Solution: CORS enable.

Planning nalla irunthathaala video-la paartha maadhiri 15 mins-la full project demo panna mudinjithu.

CODE IMPLEMENTATION:

```
# project_plan.py - Planning Phase Code
timeline = {
    'Week 1': {
        'task': 'Setup & Auth',
        'work': ['Create pocket_smart_ai folder',
                'Install Flask, JWT', 'Build login.py'],
        'status': 'Done - Video 0-4 mins'
    },
    'Week 2': {
        'task': 'Planner Modules',
        'work': ['Home Planner logic', 'Party Planner',
                'Jewelry Planner with image'],
        'status': 'Done'
    },
    'Week 3': {
        'task': 'Gemini Integration',
        'work': ['Get API key, Gemini 1.5 Flash connect',
                'Frontend dashboard cards'],
        'status': 'Done - Video 4-10 mins'
    },
    'Week 4': {
        'task': 'Testing & Demo',
        'work': ['Test with 10k budget', 'Test party 40 guests',
                'Jewelry saree upload', 'Final demo'],
        'status': 'Done - Video 10-15 mins'
    }
}
```

```
for week, details in timeline.items():
    print(f'{week}: {details["task"]} - {details["status"]}')

```

CODE EXPLANATION:

- 1. timeline Dictionary: Python dict use panni full 4 weeks plan-a oru idathula vechom. Week 1 to Week 4 varaikum enna pannanum-nu clear.
- 2. Nested Dictionary: Each week-ku task, work list status irukku. Ithu Agile board maadhiri. Video-va partha neenga Week 1 work-a first 4 mins-la pannuneenga.
- 3. Work List Array of tasks. Example: Week 1 la folder create, Flask install, login.py build. Ithula oru task missaana kooda therinjidum.
- 4. Status Tracking: 'Done - Video 0-4 mins' nu status potathu enna work eppa mudinjithu-nu track panna. Project manager-kita kaamikka easy.
- 5. For Loop: timeline.items() loop panni each week print panrathu. Sprint review meeting-la itha use pannalam.
- 6. Benefit: Intha code-a partha yarkum project eppadi plan pannom-nu puriyum. Professor-kita explanation kudukka easy. 15-min video-ku intha planning than base.
- 7. Real-Life Mapping: Video-la neenga Flask app run panni 127.0.0.1:5000 login panna athu Week 3 & 4 work. Itha plan pannama panna mudiyathu.